



AI Driven SQA: Manual & Automation Testing

Assignment 03

Module: 03

Topic: Software Testing Life Cycle (STLC) with AI

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Batch: 21

Test Plan: Ghorer Bazar Online Shopping Website

1. Document Control

Item	Details
Project	Ghorer Bazar Online Shopping Website
Website	ghorerbazar.com
Document type	Software Test Plan
Prepared by	SQA Student
Target users	Online shoppers in Bangladesh
Currency	Bangladeshi Taka, or Tk
Primary languages	Bengali and English

The landing page currently exposes an “All Product” catalog containing grocery and food products such as mustard oil, ghee, honey, dates, tea, spices, rice, lentils, pickles, and nuts. Product cards display product names and regular/sale prices in Tk.

2. Objectives

The objectives of testing are to ensure that

- The users can find the products,
- The information about them, including images, variants, availability, and prices, is up-to-date
- The quantity and products in the cart are updated
- The checkout process is appropriate for the customer’s data and delivery method
- Orders are created only after successful validation and payment confirmation.
- The website works reliably on supported browsers, devices, and network conditions.
- Bengali and English content are usable and displayed correctly.
- Security, privacy, accessibility, and performance risks are identified before release.

3. Scope

In scope

- Home page and navigation.
- Product listing and product details.
- Categories and collections.
- Search, filtering, and sorting.
- Product variants and quantity selection.
- Cart and mini-cart.
- Login, registration, logout, and password recovery.

- Address and delivery selection.
- Payment methods.
- Order confirmation and order history.
- Email/SMS notifications, if implemented.
- Responsive UI and browser compatibility.
- Accessibility, security, performance, and basic SEO checks.
- Admin/API integration validation, where test access is available.

Out of scope

- Testing the accuracy or quality of the physical food products.
- Testing third-party payment-provider infrastructure beyond the website integration.
- Penetration testing of hosting infrastructure.
- Warehouse, inventory, and delivery operations outside the system interface.
- Production data modification.
- Destructive load testing against the live website.

4. Quality Risks

Risk area	Reason
Checkout and payment	Failure can cause lost sales, duplicate charges, or incorrect orders
Cart totals and pricing	Incorrect totals directly affect customer trust and financial reconciliation
Order creation	Orders must not be lost, duplicated, or created with incorrect details
Product availability and variants	Customers may order an unavailable or incorrect package size
Delivery fee and address validation	Incorrect delivery information can cause failed fulfillment
Authentication and account data	Account or personal-data exposure is high impact
Mobile responsiveness	A large proportion of online shopping traffic may be mobile
Search, filter, and sort	Impacts discoverability but usually does not block checkout

5. Test Strategy

Test levels

- **Unit testing:** Testing business rules like how price calculated, how discounts work, if stock is available and if addresses are correct.
- **API and integration testing:** Checking how different parts like products, carts, customers orders, inventory, delivery, payments and notifications work together.
- **System testing:** Testing user journeys that happen in a browser.

- **End-to-end testing:** Testing the process from browsing a product to adding it to a cart checking out paying, getting a confirmation and seeing the order, in history.
- **Regression testing:** Making sure that new changes or updates do not break the things that already work.
- **User acceptance testing:** Letting business users test and make sure that the product, pricing, delivery and order processes work as expected.
- **Production smoke testing:** Quick checks after a deployment to make sure everything is working without causing any damage.

Test techniques

- Requirements-based testing.
- Equivalence partitioning.
- Boundary-value analysis.
- Decision-table testing.
- State-transition testing.
- Exploratory testing.
- Risk-based regression testing.
- Negative testing and error-message validation.
- Accessibility and usability review.

Test types

- Functional testing.
- UI and visual testing.
- Responsive testing.
- Cross-browser testing.
- Compatibility testing.
- API testing.
- Integration testing.
- Database/data-integrity testing.
- Performance testing.
- Security testing.
- Accessibility testing.
- Localization testing.
- Recovery and resilience testing.
- Analytics and tracking validation.

6. Functional Test Coverage

6.1 Home page and navigation

Verify that:

- The website loads successfully over HTTPS.
- Logo, menus, banners, promotional areas, and footer links work.
- Navigation menus open and close correctly on desktop and mobile.
- Category and collection links route to the correct pages.
- Product cards are displayed without broken images or missing text.
- Sale and regular prices are formatted correctly.
- Product names support both Bengali and English text where applicable.

- Browser Back, Forward, Refresh, and direct URL navigation work.
- Broken links, incorrect redirects, and unexpected 404 pages are not present.

6.2 Product listing

The current catalog includes multiple package sizes and product types—for example, 500 g and 1 kg honey, several date variants, and different sizes of Lachcha Semai—so variant and unit validation should receive high priority.

Test the following:

- Product listing loads with the correct number of products.
- Product name, image, price, sale price, unit, and availability are correct.
- Regular price and sale price are not reversed.
- Product cards link to the correct product details.
- Pagination or infinite scrolling works correctly.
- Product ordering remains stable when navigating between pages.
- Out-of-stock products are correctly labeled.
- Products cannot be purchased above available inventory.
- Duplicate products do not appear unexpectedly.
- Bengali characters, punctuation, and product units render correctly.

6.3 Search, filter, and sort

Verify that:

- Exact product-name searches return the correct product.
- Bengali, English, mixed-language, and case-insensitive searches work as intended.
- Leading/trailing spaces are handled correctly.
- Empty searches display a useful state.
- No-result searches display a clear message and recovery option.
- Special characters and potentially malicious search strings are handled safely.
- Clearing one filter does not unintentionally clear all filters.
- Price-range boundaries work correctly.
- Sorting by price, newest, popularity, or relevance produces the expected order.
- Search and filter states persist appropriately during navigation.
- Search performance remains acceptable for large catalogs.

6.4 Cart

Test:

- Adding one product.
- Adding multiple products.
- Adding the same product repeatedly.
- Adding different variants of the same product.
- Updating quantity using plus/minus controls.
- Entering a quantity manually.
- Removing an item.
- Clearing the cart.
- Recovering a cart after refresh.
- Cart behavior after login and logout.
- Cart behavior across multiple tabs.

- Stock changes after an item has been added.
- Empty-cart state.

6.5 Login, registration, and account

Verify:

- Registration with valid data.
- Required-field validation.
- Duplicate email or phone-number handling.
- Invalid email and phone formats.
- Password policy enforcement.
- Password masking and show/hide behavior.
- Login with valid and invalid credentials.
- Account lockout or throttling after repeated failures.
- Forgot-password flow.
- Password-reset token expiration and reuse prevention.
- Logout from all relevant pages.
- Session expiration.
- Account details update.
- Address-book management.
- Order-history visibility.
- Unauthorized access to another customer's account or order.
- Guest-to-account conversion after checkout, if supported.

6.6 Checkout and delivery

Verify:

- Checkout can be started from the cart.
- Required customer and delivery fields are clearly identified.
- Bengali and English names are accepted appropriately.
- Phone-number validation supports the intended Bangladesh format.
- Address fields handle valid and invalid values.
- District, city, area, and postcode dependencies work correctly.
- Saved addresses can be selected, edited, and deleted.
- Delivery options and delivery fees are correct.
- Delivery fee changes when the destination or order total changes.
- Unsupported delivery areas are blocked with a clear explanation.
- Billing and shipping addresses can be the same or different where applicable.
- Order summary matches the cart.

6.7 Payment

Verify:

- Each supported payment method is shown properly on the screen.
- When a payment succeeds one and one order is created.
- If a payment fails no confirmed order is made.
- If a payment is cancelled the user is returned to the website without issues.
- Payment timeouts are managed without errors.
- If the network fails during payment no duplicate orders are created.
- Refreshing the browser during payment does not cause problems.

- The payment amount matches the final order total exactly.
- The currency and decimal points are correct in all cases.
- The payment status stays in sync, between the payment provider and the order system.
- Delayed payment callbacks are processed correctly.
- Repeated callbacks do not create orders or duplicate payments.
- If refunds or cancellations are supported their status is shown accurately
- The website does not Log sensitive payment details.
- Users cannot change the payment amount by manipulating the client side.

7. Non-Functional Test Plan

Performance

- Home-page load time.
- Product-list load time.
- Search response time.
- Add-to-cart response time.
- Checkout page load time.
- Payment initiation time.
- API response time and error rate.

Security

Perform authorized security testing for:

- HTTPS and certificate configuration.
- Authentication and session management.
- Broken access control.
- Customer/order ID tampering.
- SQL injection.
- Cross-site scripting.
- Cross-site request forgery.
- Open redirects.
- Clickjacking protection.
- Rate limiting on login, password reset, search, and checkout endpoints.
- Secure cookies and appropriate cookie flags.

8. Representative End-to-End Scenarios

Scenario A: Guest purchase

- Open the website.
- Search for “honey” or its Bengali equivalent.
- Open a 1 kg honey product.
- Verify product information and price.
- Add the product to the cart.
- Change quantity to two.
- Confirm subtotal and total.

- Proceed as a guest.
- Enter valid customer and delivery information.
- Select a delivery option.
- Complete payment using sandbox credentials.
- Verify one order confirmation.
- Verify the order appears in the configured notification or order-management system.

Scenario B: Failed payment recovery

- Add a product to the cart.
- Enter valid checkout information.
- Use a sandbox payment that fails or is cancelled.
- Confirm that no confirmed order is created.
- Return to checkout or cart.
- Retry with a successful payment.
- Verify exactly one successful order and correct inventory behavior.

Scenario C: Stock and price change

- Add a product to the cart.
- Change its stock or price through the authorized test-admin process.
- Return to the cart or checkout.
- Confirm that the customer sees the current state.
- Verify that checkout cannot proceed with an unavailable product or stale price.

Scenario D: Mobile Bengali checkout

- Open the website on a mobile viewport.
- Switch to Bengali, if supported.
- Search for a Bengali product name.
- Add a product and open the cart.
- Complete checkout using touch input and the mobile keyboard.
- Verify no fields or buttons are hidden.
- Confirm the order summary and confirmation text are readable.

9. Test Data

Prepare controlled data for:

- Active, inactive, discounted, and out-of-stock products.
- Products with one and multiple variants.
- Products with minimum and maximum purchase quantities.
- Valid and invalid Bangladesh phone numbers.
- Valid and invalid email addresses.
- Multiple delivery districts and unsupported areas.
- Guest and registered customers.
- New, used, expired, and restricted coupons.
- Successful, failed, cancelled, and delayed payment transactions.
- Boundary prices, quantities, discount values, and delivery fees.

10. Environment and Tools

Browsers

Browser	Versions	OS
Google Chrome	Latest & Latest-1	Windows 11, macOS
Mozilla Firefox	Latest	Windows 11, macOS
Safari	Latest	macOS, iOS
Microsoft Edge	Latest	Windows 11
Chrome (Mobile)	Latest	Android 12+
Safari (Mobile)	Latest	iOS 16+

Tools

- Test management: Jira, TestRail, Zephyr, or equivalent.
- UI automation: Playwright or Selenium.
- API testing: Postman, Newman, or REST Assured.
- Performance: k6, JMeter, or Gatling.
- Accessibility: axe-core, Lighthouse, and manual screen-reader testing.

11. Entry and Exit Criteria

Entry criteria

Testing begins when:

- A testable build is deployed.
- Requirements and acceptance criteria are available.
- Critical user journeys are defined.
- Test data and test accounts are prepared.
- Sandbox payment credentials are available.
- Required integrations are reachable.
- No known blocker prevents basic navigation.
- Build version and deployment notes are documented.

Exit criteria

Testing may be completed when:

- All test cases pass.
- At least 95% of planned P1 cases pass, unless formally waived.
- No open Critical or Blocker defects remain.
- High-severity defects have an approved resolution or workaround.
- Checkout, payment, order creation, inventory, and notification flows pass end to end.
- Supported browsers and mobile layouts pass regression.
- Security and accessibility risks are reviewed.

12. Defect Classification

Severity	Definition	Example
Critical	Security breach, data loss, or complete business failure	Payment succeeds but no order is created
Blocker	Testing or a primary user journey cannot continue	Website or checkout is unavailable
High	Major function fails with no practical workaround	Incorrect total or duplicate order
Medium	Important issue with a workaround	Filter returns inconsistent results
Low	Cosmetic or low-impact usability issue	Minor alignment or typography defect

Each defect should include:

- Defect ID and title.
- Environment, build, browser, and device.
- Preconditions.
- Reproduction steps.
- Expected result.
- Actual result.
- Severity and priority.
- Business impact.

Prepare a Test Checklist for Ghorer Bazar Online Shopping Website

1. Login Checklist

20 validation points covering UI, field validation, security, session management, functional flow, usability, compatibility, and performance.

Test Area	Validation Point	Expected Result
UI	Login page loads with fields for email/phone and password, "Sign In" button, and "Forgot Password" link	All elements render correctly and are visible without layout issues on desktop and mobile
Field Validation	Submit the login form with both fields empty	Inline validation errors are shown for required fields; no request is sent to the server
Field Validation	Enter a registered email/phone with the correct password	User is logged in successfully and redirected to the homepage or previous page

Field Validation	Enter a registered email/phone with an incorrect password	Generic error message (e.g., "Invalid credentials") is shown; account/email existence is not revealed
Field Validation	Enter an unregistered email/phone number	Generic "Invalid credentials" error is shown, not a specific "user not found" message
Field Validation	Enter an invalid email format (e.g., "abc@") or invalid phone format	Format validation error is displayed before submission
Field Validation	Enter password with leading/trailing spaces	Login either trims spaces correctly or rejects with a clear message, consistent with registration rules
Security	Attempt login with SQL injection or script payloads in the username/password fields	Input is sanitized/rejected; no script executes and no database error is exposed
Security	Inspect password field on submission	Password is masked on screen and transmitted only over HTTPS (never in plain query string)
Security	Attempt multiple failed logins in a row (5+)	Account lockout, CAPTCHA, or rate-limiting is triggered to prevent brute-force attacks
Security	Check password field behavior	"Show/Hide password" toggle (if present) works correctly and does not break masking on submit
Session Management	Successfully log in and check header/menu	Header updates to show logged-in state (e.g., user name/"Hello [Name]") instead of "Sign In"
Session Management	Log in, then close and reopen the browser	Session persists (if "Remember Me" selected) or requires re-login (if not), matching intended behavior
Session Management	Log in on two different browsers/devices with the same account	Both sessions behave as intended (concurrent sessions allowed or single-session policy enforced)
Session Management	Click "Logout"	User is logged out, session/token is invalidated, and cart/wishlist state is handled per requirements
Session Management	After logout, click browser "Back" button	User is not able to view authenticated/account pages from cache; redirected to login if needed

Functional	Click "Forgot Password" link	User is redirected to password reset flow (email/OTP based) and can successfully reset the password
Functional	Log in, add items to cart as guest, then log in	Guest cart items merge correctly with the logged-in user's cart (no items lost or duplicated)
Functional	Click "Register" link from the login page	User is redirected to the registration page correctly
Usability	Press "Enter" key after typing credentials	Form submits via Enter key, not just by mouse click on the button

2. Registration Checklist

20 validation points covering UI, field validation, security, OTP/verification, functional flow, usability, and compatibility.

Test Area	Validation Point	Expected Result
UI	Registration page loads with fields for name, email/phone, password, confirm password, and "Register" button	All fields and labels render correctly on desktop and mobile without overlap
Field Validation	Submit the registration form with all fields empty	Inline required-field errors are shown for each mandatory field; form is not submitted
Field Validation	Enter a name with numbers or special characters	Field either rejects invalid characters or shows a validation message, per business rules
Field Validation	Enter an invalid email format (e.g., missing "@" or domain)	Real-time or on-submit validation error is displayed
Field Validation	Enter a phone number with incorrect length or non-Bangladesh format	Validation error shown; only correctly formatted local numbers are accepted
Field Validation	Enter a password shorter than the minimum required length	Password strength/length error is displayed with the minimum requirement stated
Field Validation	Enter a weak password (e.g., "12345") if strength rules apply	Password strength indicator/error guides the user to a stronger password
Field Validation	Enter mismatched values in "Password" and "Confirm Password"	Clear error message indicates the passwords do not match; form is not submitted
Field Validation	Register using an email/phone	Clear "account already exists"

	number that already exists in the system	message is shown, with a link/prompt to log in instead
Field Validation	Enter valid data in all fields and submit	Account is created successfully and user is redirected/logged in or asked to verify
Security	Attempt to inject script/SQL payloads into name, email, or password fields	Input is sanitized; no script executes and no raw database/server error is shown
Security	Inspect password and confirm-password fields	Both fields mask input by default; optional show/hide toggle works correctly
Security	Submit the registration form multiple times rapidly (double-click / bot simulation)	Duplicate accounts are not created; duplicate submissions are prevented or debounced
Verification	Complete registration where OTP/email verification is required	Verification code/link is sent promptly and successfully activates the account when entered
Verification	Enter an incorrect OTP during verification	Clear error is shown and the user can retry within an allowed attempt limit
Verification	Let the OTP expire before entering it	Expired-OTP message is shown with an option to resend a new code
Verification	Click "Resend OTP/Code"	New code is sent and old code is invalidated; resend is rate-limited to prevent abuse
Functional	Complete registration and check for a confirmation/welcome email or SMS	Confirmation message is received with correct account details and no broken links
Functional	Register successfully and check the account/profile section	Entered details (name, email, phone) are saved and displayed accurately in the profile
Functional	Click the "Login" link from the registration page	User is redirected to the login page correctly

3. Checkout Checklist

20 validation points covering UI, cart summary, field validation, coupons/combos, payment, order confirmation, data integrity, edge cases, security, and compatibility.

Test Area	Validation Point	Expected Result
UI	Navigate to checkout via "Buy Now" and via "Cart → Checkout"	Both paths land on the same checkout page with correctly pre-filled cart/product details
Cart Summary	Review the order summary on the checkout page	Product names, quantities, unit prices, and subtotal exactly match what was added to the cart
Field Validation	Submit checkout with mandatory fields (name, phone, address) left	Inline validation errors are shown for each missing required field;

	empty	order is not placed
Field Validation	Enter an invalid phone number format in the delivery contact field	Validation error is shown before the order can be submitted
Field Validation	Select a delivery area/district from the dropdown	Delivery charge updates automatically and correctly based on the selected area
Field Validation	Leave the delivery address field with only whitespace characters	Field is treated as empty and validation error is triggered (whitespace-only is not accepted)
Functional	Increase/decrease item quantity from the checkout/cart summary	Line total and order grand total recalculate immediately and correctly
Functional	Remove an item directly from the checkout page (if supported)	Item is removed, totals update, and checkout remains valid or redirects to cart if empty
Coupon/Promo	Apply a valid, active coupon code	Discount is correctly calculated and deducted from the order total; confirmation message shown
Coupon/Promo	Apply an expired or invalid coupon code	Clear error message is shown; discount is not applied and total remains unchanged
Coupon/Promo	Apply a coupon, then remove it before placing the order	Discount is removed and total reverts correctly to the original amount
Coupon/Promo	Attempt to apply the same single-use coupon twice (two separate orders)	System rejects the second use with an appropriate "already used" message
Combo Pricing	Add a combo deal item to the cart and proceed to checkout	Combo price and "Save X%" discount are reflected accurately in the checkout total
Payment	Select "Cash on Delivery" and place the order	Order is placed successfully with COD selected; confirmation page/order ID is displayed
Payment	Select an online/mobile payment method (e.g., bKash/Nagad/card) and complete payment	User is redirected to the payment gateway, and a successful payment confirms the order correctly
Payment	Cancel the payment midway on the gateway page	User is returned to checkout/cart with the order left unplaced (not charged, not falsely confirmed)
Payment	Simulate a payment failure (e.g., insufficient funds/sandbox failure response)	Clear failure message is shown and the user can retry payment without duplicate order creation
Order Confirmation	Successfully place an order	Order confirmation page displays order ID, items, total, delivery address, and estimated delivery
Order Confirmation	Check for confirmation email/SMS after order placement	Confirmation notification is received promptly with accurate order details
Data Integrity	Place an order for an in-stock item and check inventory afterward	Stock quantity is correctly decremented in the system after successful checkout

